

Exp 36-Scatter Plot for Three Groups Weight vs Height

Aim

To write a Python program using Matplotlib to draw a scatter plot for three different groups comparing their weights and heights.

Procedure

1. Import matplotlib.pyplot.
2. Define weight and height data for three groups.
3. Plot each group using a different color and marker.
4. Add X-axis and Y-axis labels.
5. Add title and legend.
6. Display the scatter plot.

Code

```
36.py - C:/Users/subha/OneDrive/Desktop/Query processing/36.py (3.12.10)
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt

# Group 1 data
weight1 = [55, 58, 60, 62, 64, 66, 68, 70]
height1 = [115, 125, 135, 150, 155, 165, 175, 180]

# Group 2 data
weight2 = [56, 59, 61, 63, 65, 67, 69, 71]
height2 = [120, 130, 145, 160, 170, 185, 195, 220]

# Group 3 data
weight3 = [57, 60, 62, 64, 66, 68, 70, 72]
height3 = [125, 98, 152, 125, 135, 165, 235, 175]

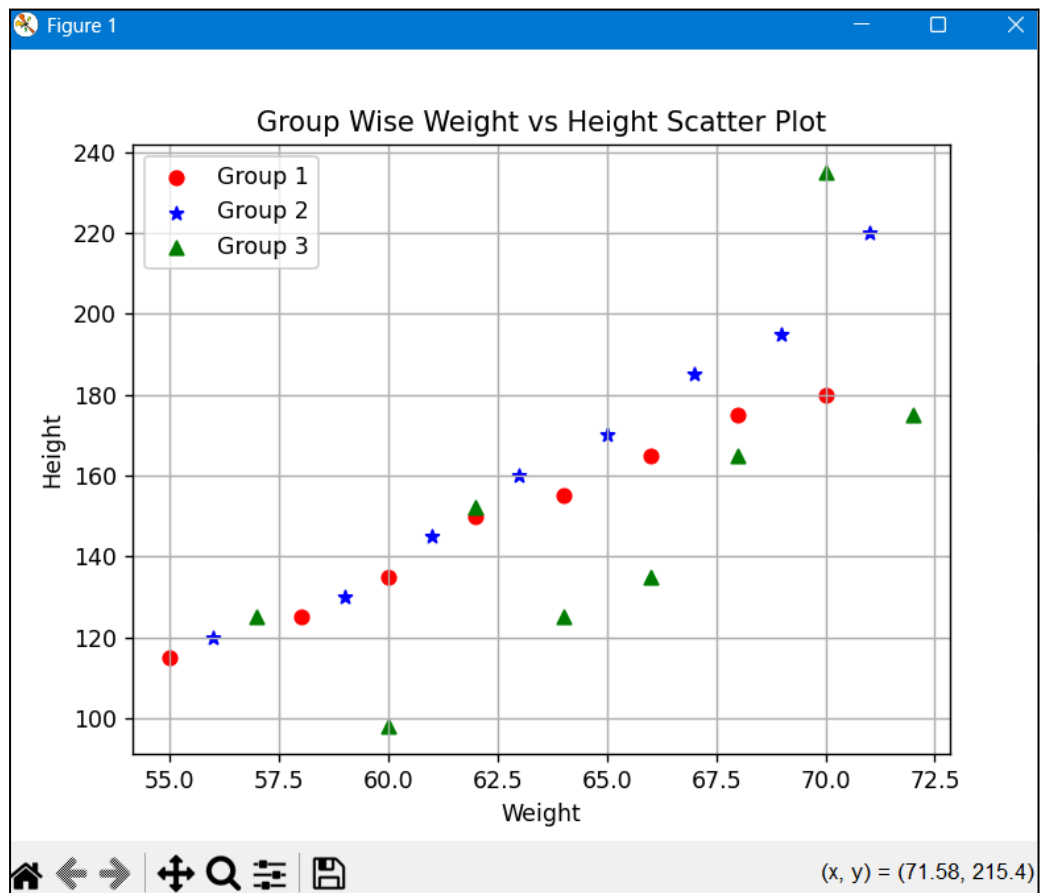
# Scatter plots
plt.scatter(weight1, height1, color='red', marker='o', label='Group 1')
plt.scatter(weight2, height2, color='blue', marker='*', label='Group 2')
plt.scatter(weight3, height3, color='green', marker='^', label='Group 3')

# Labels and title
plt.xlabel('Weight')
plt.ylabel('Height')
plt.title('Group Wise Weight vs Height Scatter Plot')

# Legend and grid
plt.legend()
plt.grid(True)

# Display plot
plt.show()
```

Output



Result

Thus, the Python program successfully creates a group-wise scatter plot comparing weight and height for three groups.

Exp 37- Create a DataFrame from a Dictionary

Aim

To write a Python program using Pandas to create a DataFrame from a dictionary and display it.

Procedure

1. Import the Pandas library.
2. Create a dictionary containing X, Y, and Z values.
3. Convert the dictionary into a DataFrame using `pd.DataFrame()`.
4. Display the DataFrame.

Code

```
37.py - C:/Users/subha/OneDrive/Desktop/Query processing/37.py (3.12.10)
File Edit Format Run Options Window Help
import pandas as pd

# Create dictionary
data = {
    'X': [78, 85, 96, 80, 86],
    'Y': [84, 94, 89, 83, 86],
    'Z': [86, 97, 96, 72, 83]
}

# Create DataFrame
df = pd.DataFrame(data)

# Display DataFrame
print(df)
```

Output

	X	Y	Z
0	78	84	86
1	85	94	97
2	96	89	96
3	80	83	72
4	86	86	83

Result

Thus, the Pandas program successfully creates and displays a DataFrame from a dictionary.

Exp 38- DataFrame with Specified Index Labels

Aim

To write a Python program using Pandas to create and display a DataFrame from a dictionary with specified index labels.

Procedure

1. Import the Pandas library and NumPy.
2. Create a dictionary containing name, score, attempts, and qualify.
3. Create a list of index labels.
4. Pass the dictionary and labels to `pd.DataFrame()`.
5. Display the DataFrame.

Code

```
38.py - C:/Users/subha/OneDrive/Desktop/Query processing/38.py (3.12.10)
File Edit Format Run Options Window Help
import pandas as pd
import numpy as np

# Dictionary data
exam_data = {
    'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily',
            'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],
    'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],
    'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
    'qualify': ['yes', 'no', 'yes', 'no', 'no',
               'yes', 'yes', 'no', 'no', 'yes']
}

# Index labels
labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

# Create DataFrame
df = pd.DataFrame(exam_data, index=labels)

# Display DataFrame
print(df)
```

Output

	name	score	attempts	qualify
a	Anastasia	12.5	1	yes
b	Dima	9.0	3	no
c	Katherine	16.5	2	yes
d	James	NaN	3	no
e	Emily	9.0	2	no
f	Michael	20.0	3	yes
g	Matthew	14.5	1	yes
h	Laura	NaN	1	no
i	Kevin	8.0	2	no
j	Jonas	19.0	1	yes

Result

Thus, the Pandas program successfully creates and displays a DataFrame with specified index labels.